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Assignment 1: Develop an ML Model using Linear Regression
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🎯 Objective

Build a Machine Learning model using Linear Regression to predict a continuous numerical value based on one or more input features.

🏠 Suggested Project

House Price Prediction using Linear Regression

Develop a model that predicts the price of a house based on features such as:

- ** Area / Square Feet
- ** Number of Bedrooms
- ** Number of Bathrooms
- ** Number of Floors
- ** Age of House
- ** Location-related features

📋 Tasks

- 1) Understand the business problem.
- 2) Collect or download a suitable dataset.
- 3) Load the dataset using Pandas.
- 4) Perform basic data understanding:
 - head()
 - shape
 - info()
 - describe()
 - Check missing values
- 5) Perform Exploratory Data Analysis (EDA).
- 6) Identify:
 - Independent variables (X)
 - Dependent/Target variable (y)
- 7) Handle missing values and unnecessary columns.
- 8) Split the dataset into:
 - 80% Training
 - 20% Testing
- 9) Build a Linear Regression model using Scikit-learn.
- 10) Train the model using the training data.
- 11) Make predictions on the test data.
- 12) Evaluate the model using:

MAE
MSE
RMSE
RÂ² Score

- 13) Compare Actual vs Predicted values.
- 14) Visualize the regression results.
- 15) Test the model with new/unseen input data.
- 16) Write a short conclusion about model performance.

ðŸ’» Technologies

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** Python
** NumPy
** Pandas
** Matplotlib
** Seaborn
** Scikit-learn
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Step 1: Install Required Libraries
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numpy
pandas
matplotlib
seaborn
scikit-learn
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Step-2 : Create Data set (dataset.csv)
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area,bedrooms,bathrooms,age,price
1000,2,2,10,5000000
1200,2,2,8,6000000
1500,3,2,7,7500000
1800,3,3,5,9000000
2000,4,3,6,10000000
2200,4,3,4,11000000
2500,4,4,3,13000000
2800,5,4,2,15000000
3000,5,4,1,17000000
3500,5,5,1,20000000
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Step-3 : create main.py and implement logic
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import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import pickle

from sklearn.model_selection import train_test_split
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from sklearn.linear_model import LinearRegression

df = pd.read_csv("dataset.csv")
print(df)

# Understand data
print(df.head())
print(df.tail())
print(df.shape)
print(df.columns)
print(df.info())
print(df.describe())

# check missing values
print(df.isnull().sum())

# Check Duplicate records
print(df.duplicated().sum())

df = df.drop_duplicates()

print(df)

# Create EDA (understanding relationships between our features and house price.)

plt.figure(figsize=(8,5))
plt.scatter(df["area"], df["price"])

plt.xlabel("area")
plt.ylabel("price")
plt.title("Area vs House Price")

# plt.show()

# Define independent and dependent variables
# x = input and y = output

x = df[["area", "bedrooms", "bathrooms", "age"]]
y = df["price"]

# Split DataSet (80-20 formula)

x_train, x_test, y_train, y_test = train_test_split(x,
                                                    y,
                                                    test_size=0.2,
                                                    random_state=42
                                                    )

# check train and test data frame shapes
print("X_Train:", x_train.shape)
print("X_Test:", x_test.shape)

print("Y_Train:", y_train.shape)
print("Y_Test:", y_test.shape)

# Create model
model = LinearRegression()

# Model Training
model.fit(x_train, y_train)
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# Make Predications
y_predict = model.predict(x_test)
print(y_predict)

df = pd.DataFrame({
    "Actual Price": y_test,
    "Predicted Price": y_predict
})

print(df)

# Test with new data
new_house = pd.DataFrame({
    "area": [5000],
    "bedrooms": [7],
    "bathrooms": [7],
    "age": [1]
})

predicted_price = model.predict(new_house)
print("Predicated House Price : ", predicted_price)

## save model

with open("house_price_model.pkl", "wb") as file:
    pickle.dump(model, file)

print("***** Model Saved to Pickle file *****")
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